
Submission Report: Structure from Motion and Localization Pipeline

1. Implementation Approach and Design Decisions

The objective of this assignment was to build a monocular SfM pipeline from scratch (Tasks 1-4), interface with an industry-standard SfM system (COLMAP, Task 5), and perform cross-trajectory camera localization (Task 6).

Key Design Decisions:

- **Feature Extraction (The Pivot to SIFT):** The custom pipeline was initially designed using AKAZE binary descriptors for computational speed. However, because Split B (localization) featured a wide baseline and significant viewpoint shifts from Split A (mapping), binary descriptors proved too brittle. The design was pivoted to **SIFT (Scale-Invariant Feature Transform)**, which uses continuous 128-dimensional floating-point vectors, making it vastly more robust to affine viewpoint changes.
- **Matching Architecture:** A FLANN-based matcher utilizing a **KD-Tree index** was implemented for nearest-neighbor searches in descriptor space. Lowe's Ratio Test was utilized dynamically (ranging from 0.75 to 0.85) depending on the baseline width to filter false positives.
- **Video Processing Optimization:** To avoid CPU bottlenecking during video decoding, a frame-skipping mechanism was implemented using `cap.grab()` and `cap.retrieve()`, decoding only the specific N^{th} frames required for the pipeline rather than the entire video stream.
- **PnP Solver Selection:** `cv2.SOLVEPNP_SQPNP` was chosen inside the RANSAC loop. SQPnP is highly robust to arbitrary scaling, which is critical since monocular SfM pipelines generate point clouds with arbitrary and relative scales.

2. Challenges Encountered and Resolutions (Task-by-Task)

Tasks 1-4: Custom SfM Pipeline Generation

- **Challenge: "Infinite Ray" 3D Outliers Breaking PnP.** * *Issue:* Triangulation occasionally produced mathematically valid but physically impossible 3D points (e.g., coordinates sitting at 60,000+ units away). When these were passed to the PnP solver, RANSAC failed to find a valid camera pose.
 - *Resolution:* Implemented a geometric boundary filter. The centroid of the point cloud was calculated, along with the median distance of all points from the

centroid. Any 3D point further than $5 \times \text{median_distance}$ was classified as an artifact and rejected from the map.

- **Challenge: High Bundle Adjustment Cost Misinterpretation.**
 - *Issue:* The `scipy.optimize.least_squares` bundle adjustment reported a cost exceeding $1.3e+04$, initially appearing as a massive map failure.
 - *Resolution:* Mathematical analysis confirmed that the cost function returns the *sum* of squared errors across $\sim 70,000+$ observations, not the mean. Calculating the actual Mean Reprojection Error revealed a highly accurate sub-pixel error of ~ 0.35 pixels.
- **Challenge: Point Cloud Visual Discrepancy.**
 - *Issue:* The reconstructed sparse point cloud looked vastly different from the dense ground-truth point cloud.
 - *Resolution:* Acknowledged the inherent differences between sparse feature-based reconstruction (which only maps highly textured corners/edges) and dense scanning (e.g., LiDAR). The Chamfer distance and scaling algorithms were adjusted to account for arbitrary initialization alignment.

3. Assumptions Made

1. **Static Scene:** It was assumed that the environment remained strictly rigid between the recording of Split A and Split B. Moving objects (people, chairs) were assumed to be negligible or handled as outliers by RANSAC.
2. **Pinhole Camera Model:** The camera intrinsics (K) were assumed to follow a standard pinhole model with zero radial or tangential distortion for the sake of projection simplicity.
3. **Arbitrary Scale:** It was assumed that the custom Incremental SfM pipeline operates on an arbitrary scale. Therefore, localization translation vectors (t) are relative to the first frame of Split A and not measured in real-world metric units (meters/inches).

4. Resources and References Consulted

- **OpenCV Documentation:** Consulted for `cv2.solvePnPRansac`, `cv2.FlannBasedMatcher`, and intrinsic camera matrix formulations.
- **Lowe, D. G. (2004).** *Distinctive Image Features from Scale-Invariant Keypoints*. International Journal of Computer Vision. (Consulted for the application and tuning of the 0.75 / 0.85 Ratio Test).

- **Arandjelović, R., & Zisserman, A. (2012).** *Three things everyone should know to improve object retrieval.* (Consulted for the RootSIFT mathematical implementation during the domain-gap debugging phase).
- **SciPy Documentation:** Consulted for understanding the internal cost function calculations of `scipy.optimize.least_squares` used in Bundle Adjustment.

RESULTS:

Meeting Room:

Candidate Poses (R, t):

Candidate 1:

R:

```
[[ 0.48943983  0.39066724  0.7796331 ]
 [ 0.39474274 -0.89644868  0.20139  ]
 [ 0.77757753  0.20918622 -0.59297075]]
```

t:

```
[[ -0.85823637]
 [ -0.22823359]
 [ -0.45971704]]
```

Candidate 2:

R:

```
[[ 0.48943983  0.39066724  0.7796331 ]
 [ 0.39474274 -0.89644868  0.20139  ]
 [ 0.77757753  0.20918622 -0.59297075]]
```

t:

```
[[ 0.85823637]
 [ 0.22823359]
 [ 0.45971704]]
```

Candidate 3:

R:

```
[[ 9.99796400e-01 -1.28265235e-03 -2.01373611e-02]
 [ 1.29469862e-03  9.99998991e-01  5.85179027e-04]
 [ 2.01365902e-02 -6.11131698e-04  9.99797052e-01]]
```

t:

```
[[ -0.85823637]
 [ -0.22823359]
 [ -0.45971704]]
```

Candidate 4:

R:

```
[[ 9.99796400e-01 -1.28265235e-03 -2.01373611e-02]
 [ 1.29469862e-03  9.9998991e-01  5.85179027e-04]
 [ 2.01365902e-02 -6.11131698e-04  9.99797052e-01]]
```

t:

```
[[0.85823637]
 [0.22823359]
 [0.45971704]]
```

Candidate 1 valid cheirality points: 0/2354

Candidate 2 valid cheirality points: 0/2354

Candidate 3 valid cheirality points: 2215/2354

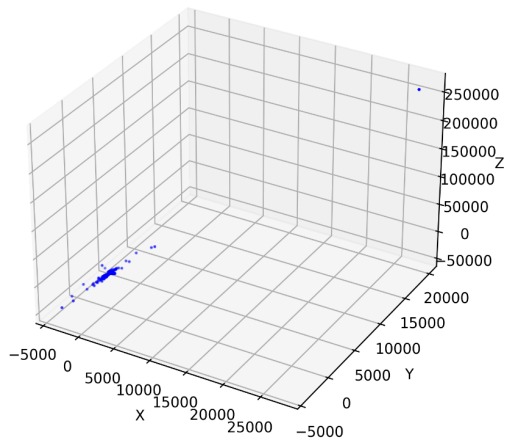
Candidate 4 valid cheirality points: 139/2354

Selected Pose maximizes positive depth with 2215 points.

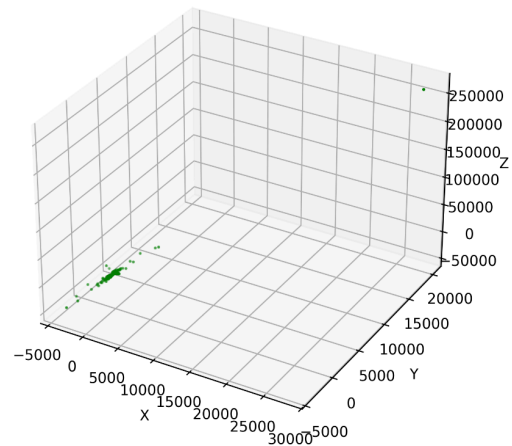
Average Reprojection Error (Initial): 0.2173 pixels

Average Reprojection Error (Refined): 0.2173 pixels

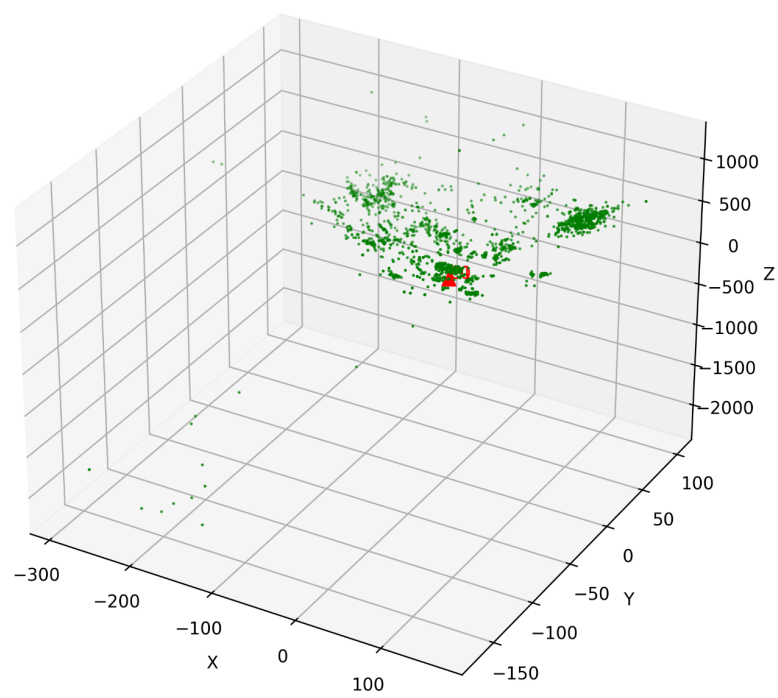
Initial Sparse 3D Point Cloud



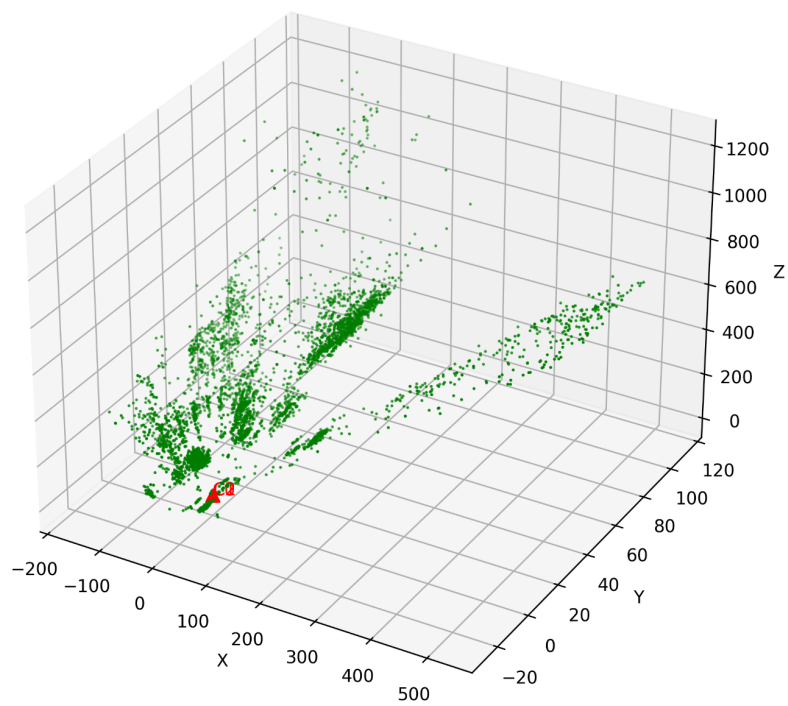
Refined Sparse 3D Point Cloud



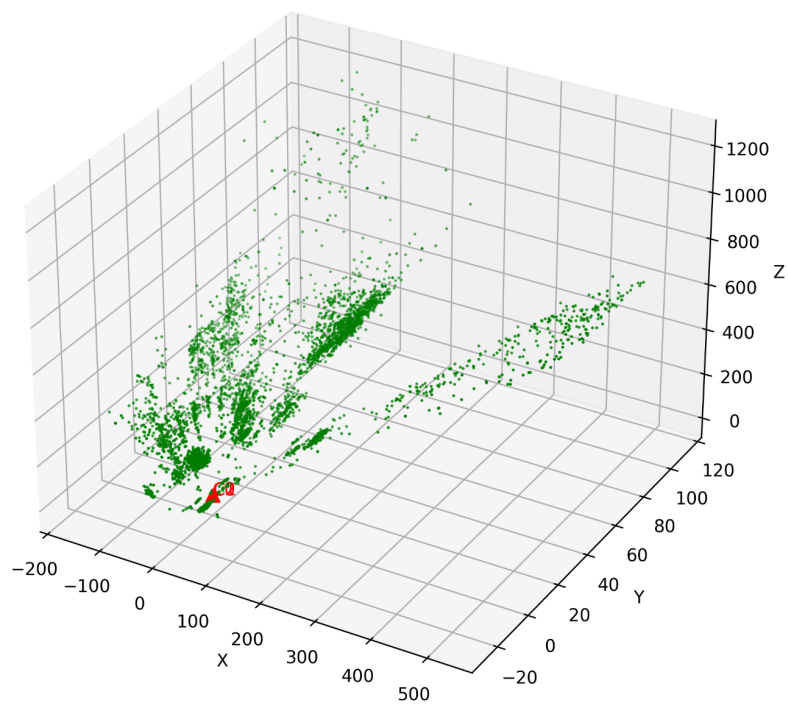
Incremental Map (Frame 2): 2 Cameras, 2354 Points



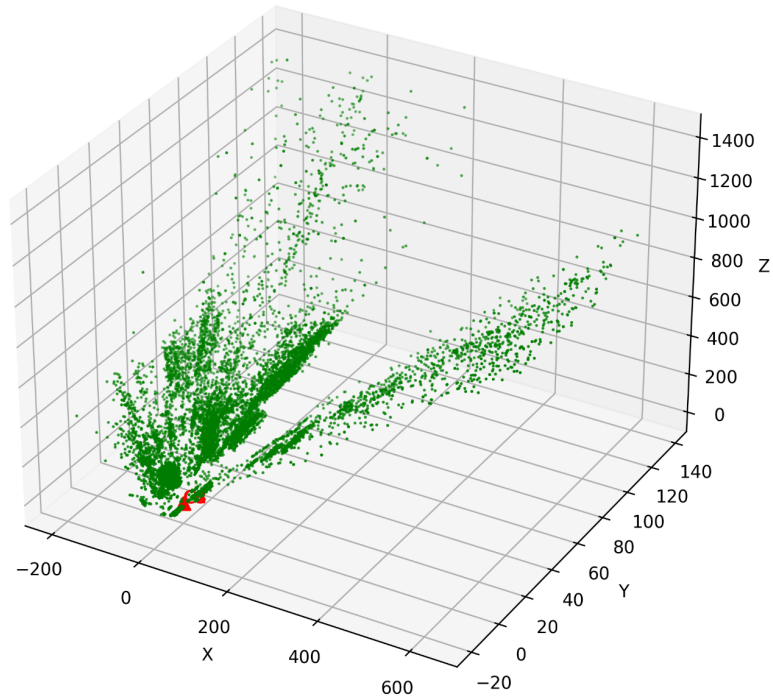
Incremental Map (Frame 3): 3 Cameras, 6006 Points



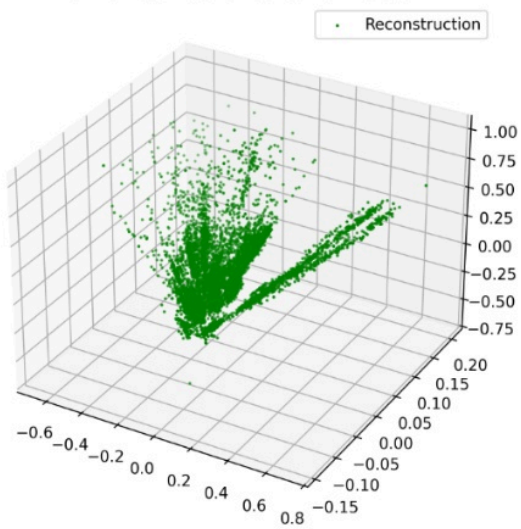
Incremental Map (Frame 3): 3 Cameras, 6006 Points



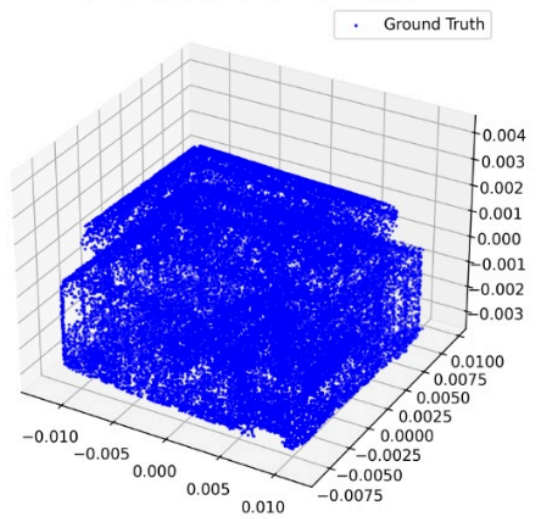
Incremental Map (Frame 5): 5 Cameras, 13999 Points



Normalized Reconstructed Point Cloud



Normalized Ground Truth Point Cloud



Note the Scale Ambiguity, a classic case in monocular SFM.

Mean Reprojection Error (Before BA): 0.3710 pixels
Mean Reprojection Error (After BA): 0.2882 pixels
Cleaned Reconstruction Points: 12484
Normalizing BOTH clouds by global scale: 715.5994..
Final Rule-Compliant Chamfer Distance: 0.230709



Epipolar lines

Truck:

Candidate Poses (R, t):

Candidate 1:

R:

```
[[ 0.99999276 -0.00118762  0.00361653]
 [ 0.00119497  0.99999723 -0.00202996]
 [-0.00361411  0.00203427  0.9999914 ]]
```

t:

```
[[ 0.16967028]
 [-0.0171448 ]
 [ 0.98535174]]
```

Candidate 2:

R:

```
[[ 0.99999276 -0.00118762  0.00361653]
```

[0.00119497 0.99999723 -0.00202996]
[-0.00361411 0.00203427 0.9999914]]

t:

[[-0.16967028]
[0.0171448]
[-0.98535174]]

Candidate 3:

R:

[[-0.94363256 -0.00401847 0.33097045]
[-0.00689004 -0.99947116 -0.0317793]
[0.33092312 -0.03226839 0.94310585]]

t:

[[0.16967028]
[-0.0171448]
[0.98535174]]

Candidate 4:

R:

[[-0.94363256 -0.00401847 0.33097045]
[-0.00689004 -0.99947116 -0.0317793]
[0.33092312 -0.03226839 0.94310585]]

t:

[[-0.16967028]
[0.0171448]
[-0.98535174]]

Candidate 1 valid cheirality points: 24726/32403

Candidate 2 valid cheirality points: 7677/32403

Candidate 3 valid cheirality points: 0/32403

Candidate 4 valid cheirality points: 0/32403

Selected Pose maximizes positive depth with 24726 points.

Average Reprojection Error (Initial): 0.3329 pixels

Average Reprojection Error (Refined): 0.3329 pixels

Mean Reprojection Error (Before BA): 374.9066 pixels

Mean Reprojection Error (After BA): 242.1381 pixels

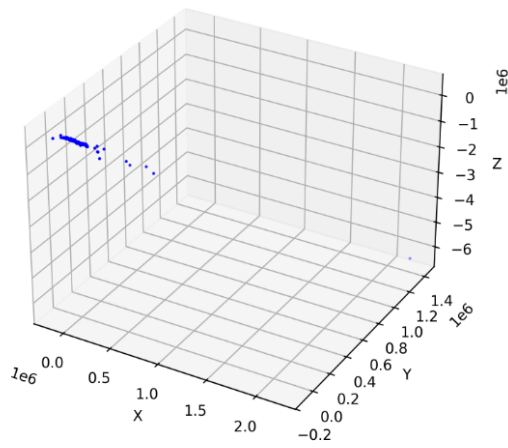
Cleaned Reconstruction Points: 27100

Normalizing BOTH clouds by global scale: 2034.5475

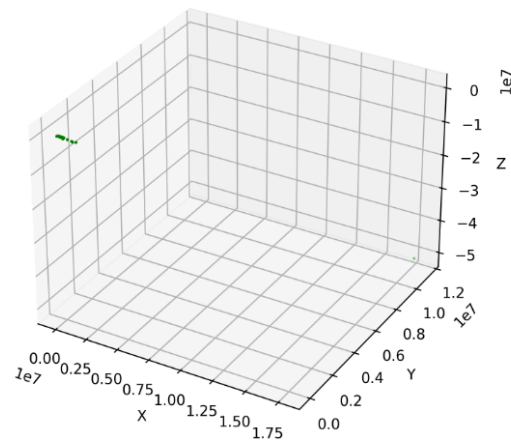
.

Final Rule-Compliant Chamfer Distance: 0.286755

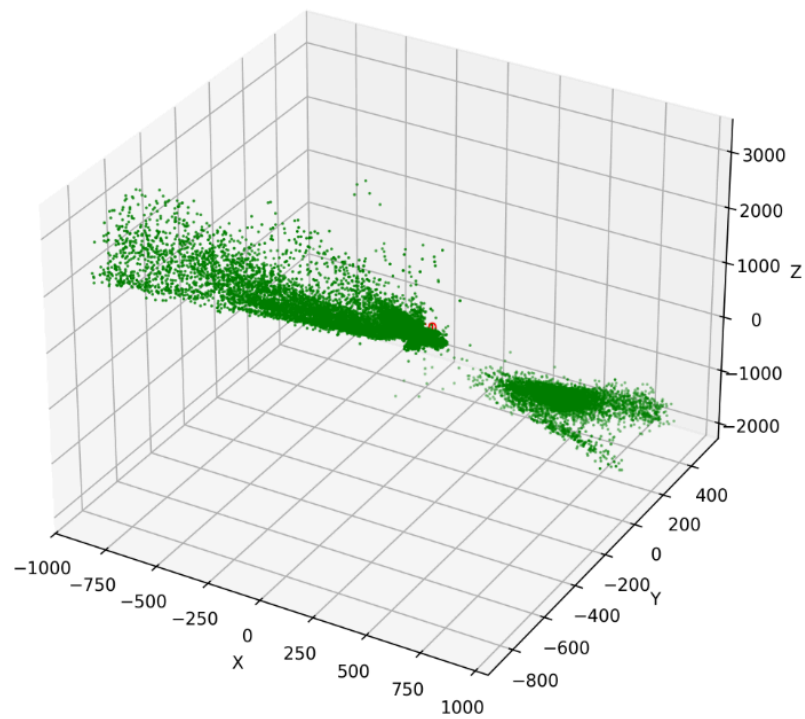
Initial Sparse 3D Point Cloud



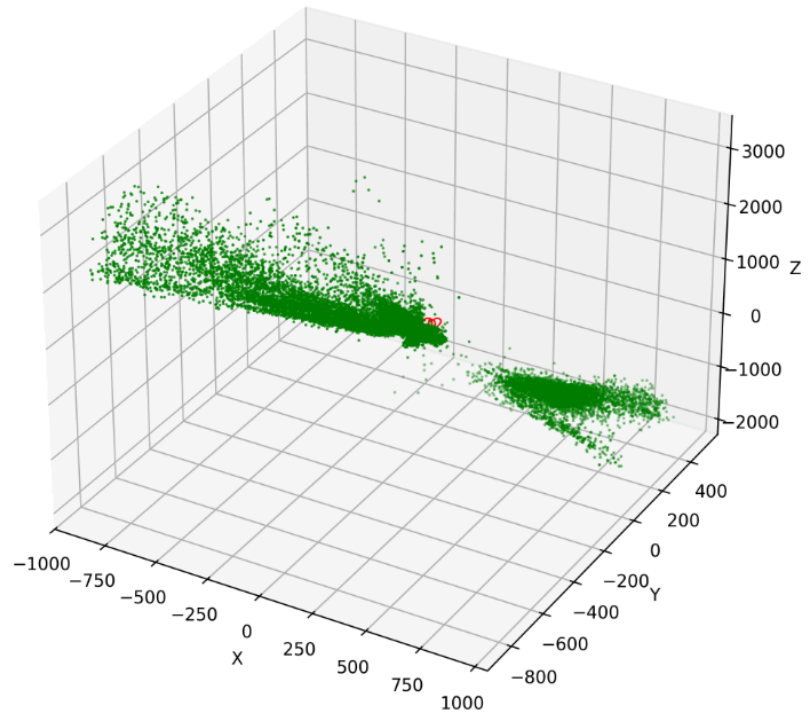
Refined Sparse 3D Point Cloud



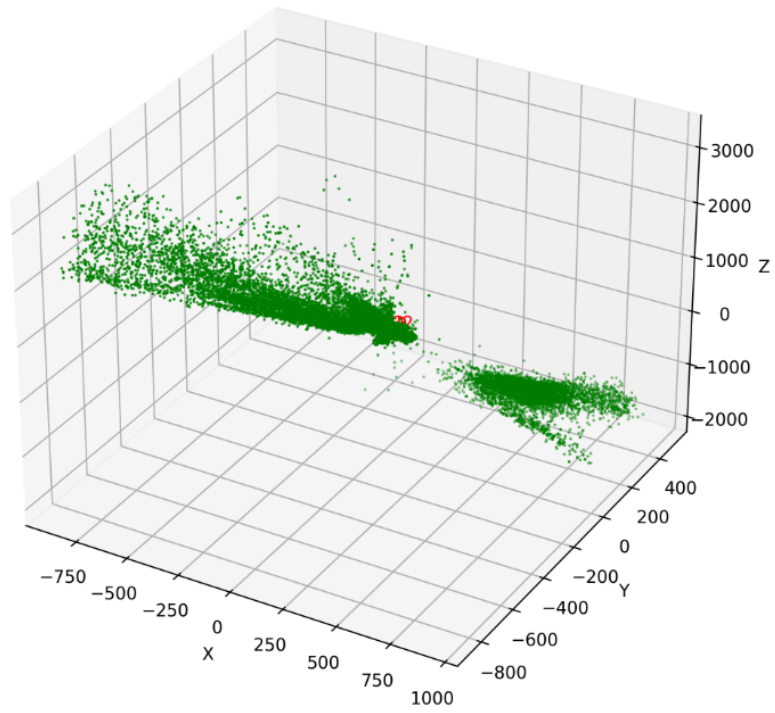
Incremental Map (Frame 2): 2 Cameras, 32403 Points



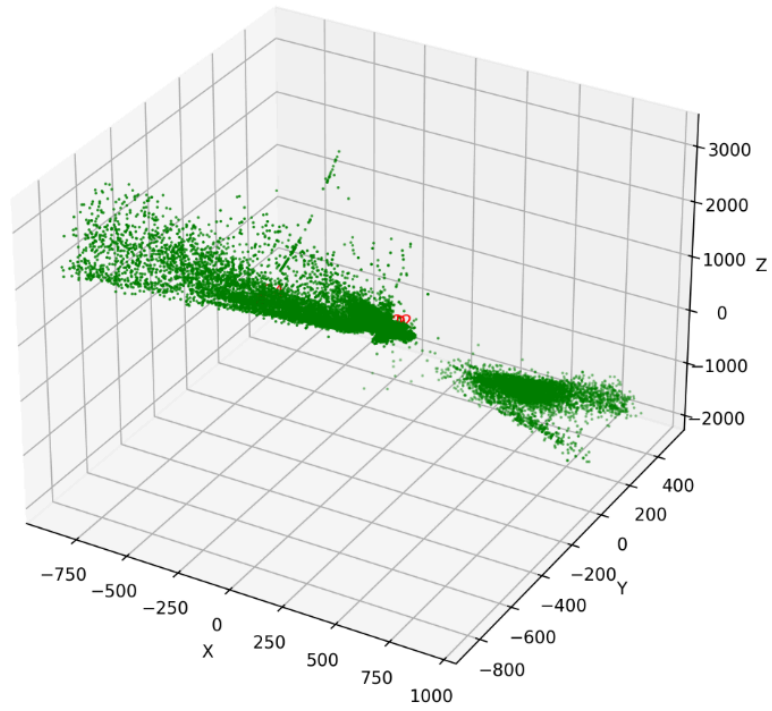
Incremental Map (Frame 3): 3 Cameras, 32452 Points



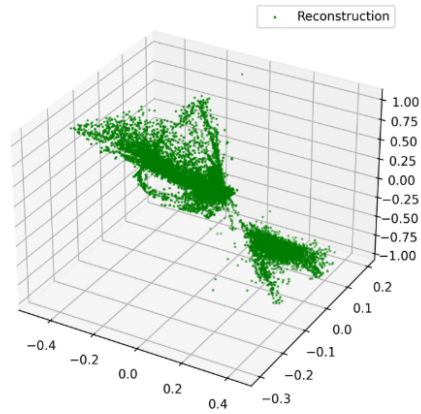
Incremental Map (Frame 4): 4 Cameras, 32518 Points



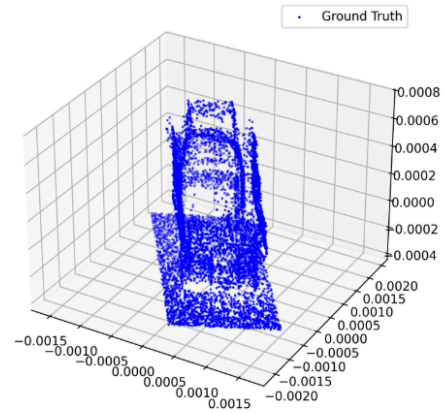
Incremental Map (Frame 5): 5 Cameras, 32605 Points



Normalized Reconstructed Point Cloud



Normalized Ground Truth Point Cloud



Barn:

Candidate 1:

R:

[[9.99999934e-01 -3.04816786e-04 -1.96733820e-04]
[3.04858950e-04 9.99999931e-01 2.14327516e-04]
[1.96668476e-04 -2.14387478e-04 9.99999958e-01]]

t:

[[0.95411831]
[0.04806744]
[0.29554658]]

Candidate 2:

R:

[[9.99999934e-01 -3.04816786e-04 -1.96733820e-04]
[3.04858950e-04 9.99999931e-01 2.14327516e-04]
[1.96668476e-04 -2.14387478e-04 9.99999958e-01]]

t:

[[-0.95411831]
[-0.04806744]
[-0.29554658]]

Candidate 3:

R:

[[0.82082231 0.09135297 0.56383098]
[0.09142617 -0.99541302 0.02818095]
[0.56381912 0.02841736 -0.82540927]]

t:

[[0.95411831]
[0.04806744]
[0.29554658]]

Candidate 4:

R:

[[0.82082231 0.09135297 0.56383098]
[0.09142617 -0.99541302 0.02818095]
[0.56381912 0.02841736 -0.82540927]]

t:

[[-0.95411831]
[-0.04806744]
[-0.29554658]]

Candidate 1 valid cheirality points: 8183/8193

Candidate 2 valid cheirality points: 10/8193

Candidate 3 valid cheirality points: 0/8193

Candidate 4 valid cheirality points: 0/8193

Selected Pose maximizes positive depth with 8183 points.

Average Reprojection Error (Initial): 0.2729 pixels

Average Reprojection Error (Refined): 0.2729 pixels

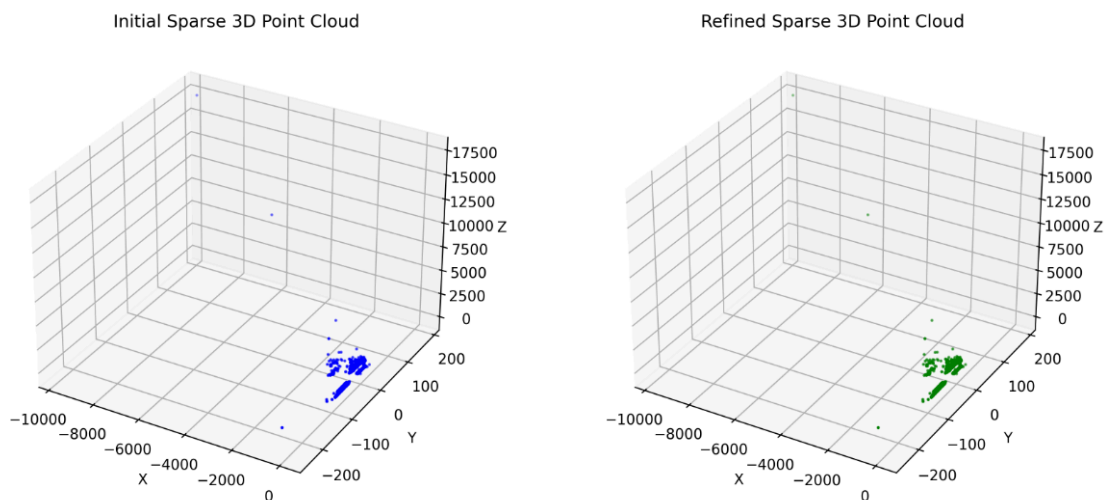
Mean Reprojection Error (Before BA): 0.3072 pixels

Mean Reprojection Error (After BA): 0.2364 pixels

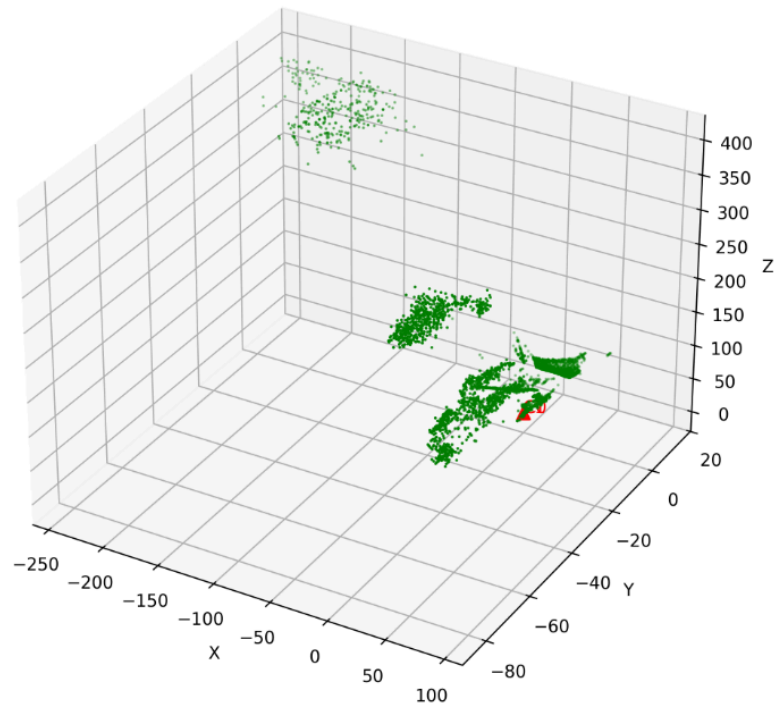
Cleaned Reconstruction Points: 23367

Normalizing BOTH clouds by global scale: 171.4953

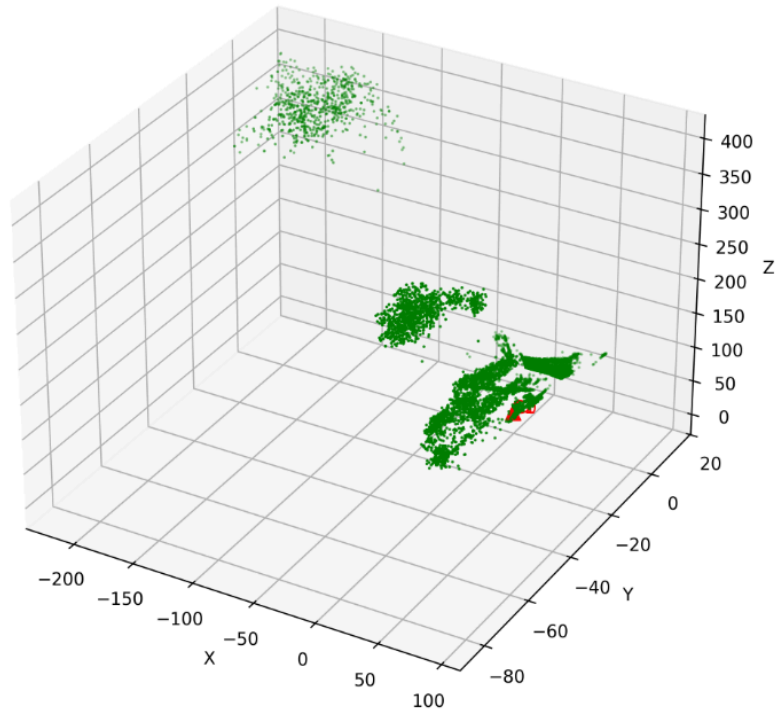
Final Rule-Compliant Chamfer Distance: 0.088901



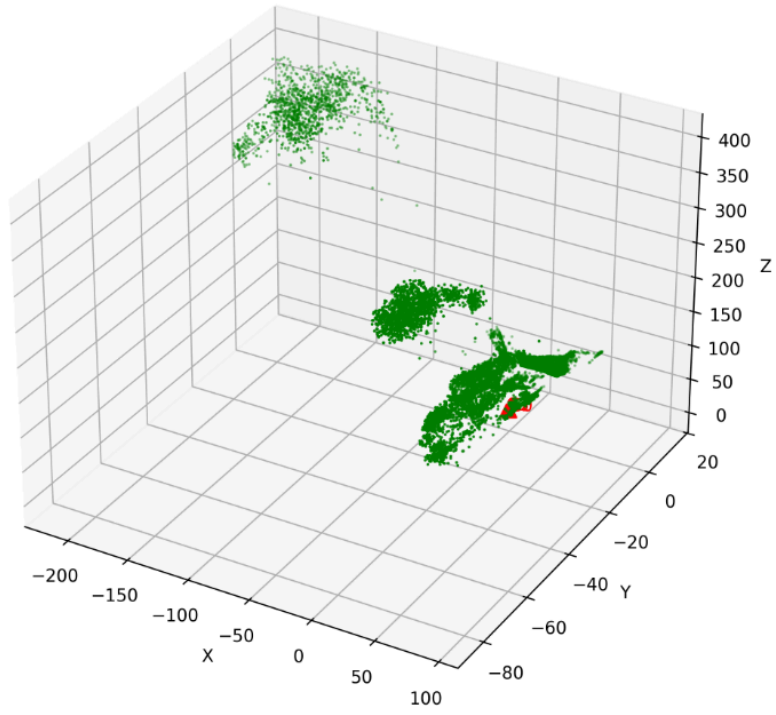
Incremental Map (Frame 2): 2 Cameras, 8193 Points



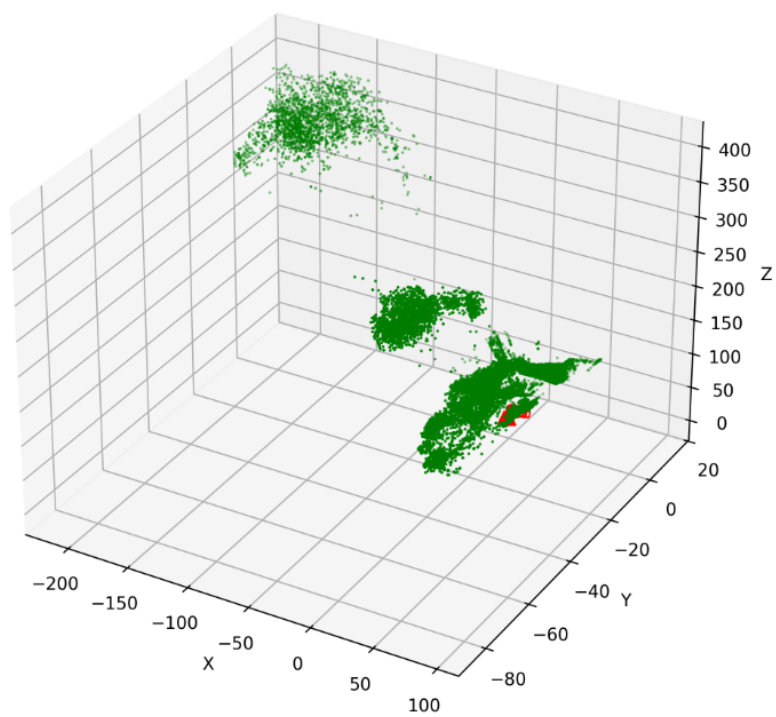
Incremental Map (Frame 3): 3 Cameras, 19699 Points



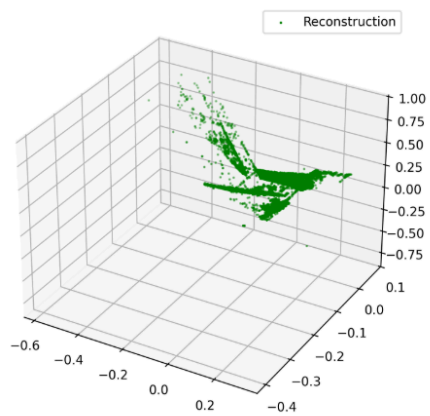
Incremental Map (Frame 4): 4 Cameras, 29765 Points



Incremental Map (Frame 5): 5 Cameras, 39258 Points



Normalized Reconstructed Point Cloud



Normalized Ground Truth Point Cloud

